

Prostate

Prostate cancer (polygenic scores, HOXB13, 8q24, repair genes), genetic PSA, benign hyperplasia and urinary symptoms, 5-alpha-reductase

Sérgio F · 02-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. Prostate cancer has a very low polygenic predisposition. Five of the six scores are below the 10th percentile: 0.9 in the genomic score of 1 million variants, 1.3 and 0.0 in the multi-population scores, 8.4 in Schumacher and 0.8 in the PHS290, the score designed to guide the screening. The GWAS loci load is at percentile 1.1. It does not have the **HOXB13 G84E** variant, none of the three high-risk alleles of the **8q24** region, nor the variants read from **CHEK2**, **BRCA2** and **ATM**. Benign hyperplasia is in the 9.3, 10.1 and 33.3 percentiles.

Unfavourable/to be considered. The genetic PSA is low: percentiles 1.3 and 6.5, and has 1 allele of **KLK3 I179T** that lowers the PSA. It is not a risk in itself, but it means that an PSA within the "normal" range may be relatively high for Sérgio F. It has some common loci with risk alleles (**TERT** and **CASC8** in homozygosity; 8q24rs6983267, NKX3-1, JAZF1 and SLC22A3 with 1 allele), but the added effect is low.

What does it mean? It is a low-risk genetic profile for the prostate. The practical point is the reading of PSA: with a low genetic PSA, the value must be followed by the trajectory from a basal value and read with a lower threshold. If finasteride or dutasteride is initiated for the hair, the measured PSA decreases even more (about half).

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Key variants** 21 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 19 of the chip, 1 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 2 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 11 PGS Catalog scores calculated locally with 2,590,246 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 510^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 1695 loci, 1391 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In cancer and hyperplasia scores, a higher percentile = more risk. In PSA scores, the percentile only says if PSA tends to be naturally higher or lower; it is not a risk.

Prostate cancer

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Prostate cancer	PGS000662 (Conti 2021, 269 variants)	221 269 (weighted coverage 76%)	1.3	LESS LIKELY	favourable
Prostate cancer	PGS003765 (Wang 2023, 451 variants)	376 451 (weighted coverage 72%)	0.0	LESS LIKELY	favourable
Prostate cancer	PGS000030 (Schumacher 2018, 147 variants)	122 147 (weighted coverage 79%)	8.4	LESS LIKELY	favourable
Prostate cancer	PGS000084 (Graff 2021, 161 variants)	147 161 (weighted coverage 94%)	69.8	TYPICAL	typical
Prostate cancer (genomic)	PGS004694 (Hu 2024, SDPR, 1.02 million variants)	991,536 1,016,708 (weighted coverage 97%)	0.9	LESS LIKELY	favourable
Age risk score (PHS290)	PGS003331 (Huynh-Le 2022, 290 variants)	242 290 (weighted coverage 81%)	0.8	LESS LIKELY	favourable

- **Risk score by age (PHS290):** Risk score over age, designed to guide the age of onset of screening.

PSA

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Genetic PSA	PGS003378 (Kachuri 2023, 128 variants)	103 128 (weighted coverage 72%)	1.3	LOW	without judgment
Genetic PSA (genomic)	PGS003379 (Kachuri 2023, PRS-CSx, 1.07 million variants)	979,034 1,071,278 (weighted coverage 62%)	6.5	LOW	without judgment

- **Genetic PSA:** It is not a risk: it says that PSA of Sérgio F tends to be naturally higher or lower, which helps to read the analysis.

Benign hyperplasia

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Prostate hyperplasia	PGS001865 (Privé 2022, 1712 variants)	1,663 1,712 (weighted coverage 97%)	9.3	LESS LIKELY	favourable
Prostate hyperplasia	PGS002076 (Privé 2022, LDpred2, 627 thousand variants)	615,841 627,341 (weighted coverage 97%)	10.1	LESS LIKELY	favourable
Prostate hyperplasia (time to event)	PGS001338 (Tanigawa 2022, 1223 variants)	961 1,222 (weighted coverage 82%)	33.3	TYPICAL	typical

4. Key variants by topic

Variants with described effect. The judgment refers to the risk of cancer and prostate hyperplasia; in PSA variants it refers to the analysis.

Large effect variants and repair genes

Rare variants of large or moderate effect. HOXB13 G84E is the main hereditary variant of prostate cancer. CHEK2, BRCA2 and ATM are DNA repair genes whose pathogenic variants mainly increase aggressive cancer. The chip reads only a few variants of these genes.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
HOXB13	rs138213197	G84E	C/C	chip	T (0 copy/s)	favourable	G84E (T na strand +): the most effective variant in hereditary prostate cancer (OR ~3-5; rare, ~0.5%)	Strong
CHEK2	rs17879961	I157T	A/A	chip	G (0 copy/s)	favourable	I157T (G in strand +): moderate variant of DNA repair (OR ~1.5-2 in prostate)	Moderate
BRCA2	rs11571833	K3326X	A/A	chip	T (0 copy/s)	favourable	K3326X: late truncation variant of BRCA2 (OR ~1.3 in several cancers)	Moderate
ATM	rs1800056	F858L	T/T	chip	C (0 copy/s)	favourable	F858L of ATM (C = Leu858; rare repair variant, uncertain effect)	Weak

Common loci of prostate cancer

Common loci of GWAS, each with small effect (OR 1.1-1.5). The 8q24 region, near the MYC, has several independent signals and is the strongest. The sum of hundreds of these loci is what polygenic scores measure.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
8q24 (2 region)	rs16901979		C/C	chip	A (0 copy/s)	favourable	8q24: Allele A is the most effective (OR ~1.5-2)	Strong
8q24 (3 region)	rs1447295		C/C	chip	A (0 copy/s)	favourable	8q24: risk allele (OR ~1.4)	Strong
8q24 (1 region)	rs4242382		G/G	chip	A (0 copy/s)	favourable	8q24: risk allele (OR ~1.4)	Strong
8q24 (POU5F1B/ MYC)	rs6983267		G/T	chip	G (1 copy/s)	unfavourable (light)	8q24: G allele increases Wnt/MYC signaling (prostate and colorectal; OR ~1.2)	Strong
MSMB	rs10993994		SNP not found in the data	SNP not found in the data (DR2<0.8 imputation)	T faux copy/s)	without data	Microseminoprotein-beta: T allele low to MSMB in semen and increases prostate cancer (OR ~1.25)	Strong
NKX3-1	rs1512268		T/C	chip	T (1 copy/s)	unfavourable (light)	Prostate tumor suppressor (NKX3-1) [GWAS Catalog: Prostate cancer, T allele, OR 1.29, p=3E-24, PMID 31562322]	Moderate
JAZF1	rs10486567		G/A	chip	G (1 copy/s)	unfavourable (light)	Locus 7p15 (prostate and diabetes type 2) [GWAS Catalog: Prostate cancer, G allele, OR 1.15, p=5E-131, PMID 37945903]	Moderate
TERT	rs2242652		G/G	chip	G (2 copy/s)	unfavourable	5p15.33 (TERT): prostate locus [GWAS Catalog: Prostate cancer, G allele, OR 1.15, p=6E-111, PMID 37945903]	Moderate

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CASC8 (8q24)	rs10086908		T/T	chip	T (2 copy/s)	unfavourable	Additional 8q24 [GWAS Catalog: Prostate cancer, T allele, OR 1.27, p=7E-14, PMID 31562322]	Weak
SLC22A3	rs9364554		C/T	chip	T (1 copy/s)	unfavourable (light)	Organic cations carrier (6q25) [GWAS Catalog: Prostate cancer, T allele, OR 1.11, p=5E-33, PMID 29892016]	Weak

Genetic PSA (which changes the analysis reading)

PSA is produced by the KLK3 gene. Variants in this region change the value of PSA in blood without changing the risk of cancer in the same proportion. Knowing the genetic PSA helps to read the analysis: who has naturally low PSA can have a "normal" value that, for itself, is already up.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
KLK3	rs17632542	I179T	T/C	chip	C (1 copy/s)	favourable (with note)	I179T of PSA (KLK3): C allele (Thr179) lowers PSA and the risk of prostate cancer (especially indolent); with it PSA reads a little lower	Strong
KLK2/ KLK3	rs2735839		A/G	chip	G (1 copy/s)	without judgment	KLK2-KLK3 region: strong genetic determinant of the value of PSA (effect allele = PSA higher) [GWAS Catalog: Prostate-specific antigen levels, allele G, beta 0.190 % increase, p=6E-37, PMID 23269536]	Strong

Androgens and 5-alpha-reductase

5-alpha-reductase type 2 (SRD5A2) converts testosterone into DHT, the active androgen of the prostate and hair follicle. CYP17A1 participates in the synthesis of androgens and AR is the receptor. The effects of these variants on the prostate are small and inconsistent.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
SRD5A2	rs523349	V89L	C/C	chip	G (0 copy/s)	neutral	V89L (G = Leu89 in the strand +): lower 5-alpha-reductase activity type 2 (less DHT); weak association with less cancer and less hyperplasia	Weak
SRD5A2	rs9282858	A49T	C/C	chip	T (0 copy/s)	favourable	A49T (T = Thr49): more activity of 5-alpha-reductase; associated with more aggressive prostate cancer (rare)	Moderate
CYP17A1	rs743572	A1/A2	A/G	chip	G (1 copy/s)	without judgment	CYP17A1 promoter (-34T>C, allele A2 = G in the strand +): androgen synthesis; weak and inconsistent association with prostate	Weak
AR (X)	rs6152		G (X)	chip	A (0 copy/s)	without judgment	StuI (E211E) of the androgen receptor: G allele is associated with more sensitivity to androgens (baldness); uncertain effect on prostate	Weak

Benign hyperplasia and inflammation

RNase L is an antiviral enzyme associated, in ancient studies, with hereditary prostate cancer. Benign hyperplasia is mainly evaluated by polygenic scores.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
RNASEL	rs486907	R462Q	C/C	imputation DR2=1.00	T (0 copy/s)	favourable	R462Q of RNase L (T = Gln462, less antiviral activity; inconsistent association with hereditary prostate cancer)	Weak

5. GWAS loci burden

Sum of all loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of each trait, with each allele oriented towards the risk (no PSA, in the sense of PSA higher), compared to the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Prostate cancer	1005 (852)	754 of 1704	845	1.1	favourable
PSA	638 (498)	486 of 996	495	12.2	without judgment
Benign hyperplasia and urinary symptoms	52 (41)	31 of 82	41	21.8	typical

6. Favourable

- **Prostate cancer very unlikely:** genomic score at percentile 0.9; multi-population scores at the 1.3 and 0.0 percentiles; Schumacher at 8.4; loci load at 1.1. Only the Graff score (161 variants) is typical (percentile 69.8). **Strong.**
- **PHS290 at percentile 0.8:** the age risk score used to decide when to start screening is very low. **Moderate.**
- **No major effect variants:** HOXB13 G84E C/C, CHEK2 I157T A/A, BRCA2 K3326X A/A and ATM F858L T/T without the risk allele. Strong (for the variants read).
- **8q24 region:** the three most effective alleles (rs16901979, rs1447295, rs4242382) are absent. **Strong.**
- **Benign hyperplasia:** scores in the percentiles 9.3, 10.1 and 33.3; loci load at percentile 21.8. **Moderate.**
- **SRD5A2 A49T absent** (C/C) and RNASEL R462Q absent (C/C). **Weak to moderate.**
- **KLK3 I179T T/C:** 1 allele associated with less prostate cancer, especially of the indolent type (with lower PSA score). **Moderate.**

7. Unfavourable

- **Low genetic PSA** (1.3 and 6.5 percentiles; PSA loci load at 12.2): does not increase the risk, but the measured PSA tends to underestimate what is happening in the prostate. A "normal" value for the population can be high for Sérgio F. **Moderate to strong.**
- **TERT rs2242652 G/G** and **CASC8 rs10086908 T/T**: homozygous for the risk allele (OR ~1.15–1.27 per allele). **Moderate / weak.**
- **Common loci with 1 risk allele**: 8q24 rs6983267 G/T, NKX3-1 T/C, JAZF1 G/A and SLC22A3 C/T. They are already counted in the scores, which are low. **Weak.**
- **No judgment**: SRD5A2 V89L C/C (typical activity of 5-alpha-reductase), CYP17A1 A/G, AR rs6152 allele G (associated with more sensitivity to androgens in the hair follicle, uncertain effect on the prostate) and KLK2/ KLK3 rs2735839 A/G.

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
DNA Damage and DNA Repair Deficit	CHEK2 I157T, BRCA2 K3326X and ATM F858L absent	Favourable
Somatic Mutation Accumulation	Prostate cancer: genomic score at percentile 0.9 and loci load at percentile 1.1	Favourable
Aberrant Transcriptional Reprogramming	HOXB13 G84E and high risk alleles of 8q24 absent; 8q24/MYC rs6983267 and NKX3-1 with 1 allele	Favourable
Hayflick's Telomeres and Limit	TERT rs2242652 G/G (prostate risk allele in homozygous)	Unfavourable (mild)
Endocrine dysregulation	Androgenic axis: SRD5A2 V89 typical, A49T absent; CYP17A1 and AR without judgment	Neutral
Chronic Inflammation	Benign hyperplasia in the 9.3–33.3 percentiles; RNASEL R462Q absent	Favourable

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Basal PSA and path tracking:** measure a basal PSA and repeat it regularly, according to the guidelines (in general from the 45–50 years). With low genetic PSA, the most useful is the rise in relation to the basal value itself, and not the population threshold. A persistent rise justifies evaluation, even with a "normal" value. If the laboratory or the doctor have access to a PSA corrected by genetics (Kachuri 2023), it is the best way to read it. *Suggestion (not applied).*
2. **If it is initiated finasteride or dutasteride for the hair** (medical decision, already mentioned in the hair plan): measure PSA before starting. At the end of 6–12 months, the measured PSA must be multiplied by 2 to compare with the previous values. Added to the low genetic PSA, without this correction PSA would be greatly underestimated. These drugs also reduce benign hyperplasia. *Suggestion (not applied).*
3. **The screening remains:** A low polygenic score reduces the probability but does not cancel it, and the chip does not read all variants of the repair genes. If PSA rises, multiparametric magnetic resonance imaging before biopsy is the usual step. *Suggestion (not applied).*
4. **Urinary symptoms:** The genetic predisposition to benign hyperplasia is low, and the tadalafil 5 mg of the plan is already indicated in the low urinary symptoms. There is nothing to add because of the genes. *Suggestion (not applied).*
5. **Lycopene at 10 mg/day** (as the plan already foresees): it arrives for the prostate, where the evidence is observational and modest. *Suggestion (not applied).*
6. **Protective lifestyle (already in the plan):** High fiber (the more, the better), crucifers and cooked tomato, omega-3 to 3.2–4 g/day, zero alcohol, regular exercise and low adiposity, which are associated with less aggressive prostate cancer. *Suggestion (not applied).*
7. **Rapamycin:** mTOR inhibition is biologically favourable in the prostate; there is nothing to change because of genes. *Suggestion (not applied).*

Not recommended: High-dose vitamin E (SELECT trial showed more prostate cancer); supplemental selenium without deficit; testosterone or androgen precursors without medical indication; interpreting a PSA under finasteride or dutasteride without multiplying by 2.

10. Limitations

- A genotyping chip with imputation reads only some variants of HOXB13, BRCA2, CHEK2 and ATM. It does not read the other pathogenic variants of these genes, nor those of BRCA1, MMR (Lynch syndrome) and other repair genes. A family history of aggressive prostate cancer, breast or ovary justifies a clinical genetic panel.
- MSMB rs10993994 and HNF1B rs7501939 are not in the data. Both enter the scores through neighboring variants.
- Polygenic scores measure relative risk. The absolute risk depends on age and family history, and a low score does not exclude the disease.
- The genetic PSA explains only a part of the variation of PSA. The prostate, inflammation and drugs also change the value.
- The load of loci GWAS sums SNPs in linkage imbalance. It is a descriptive complement and does not replace the polygenic scores.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The alleles of each location are read from the Imputed VCF. The published effect allele is compared with those alleles and, if necessary, converted to the complementary strand (only in non-palindromic SNPs). In palindromic SNPs, strand + (GWAS Catalog standard and the harmonized files of PGS Catalog) is assumed. In the loci of GWAS Catalog with protective effect, the allele of effect is exchanged by the risk allele. In the X chromosome (AR), Sérgio F is hemizygous and has only one copy.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog: PGS000662 (Conti 2021), PGS003765 (Wang 2023), PGS000030 (Schumacher 2018), PGS000084 (Graff 2021), PGS004694 (Hu 2024), PGS003331 (PHS290, Huynh-Le 2022), PGS003378 and PGS003379 (Kachuri 2023), PGS001865 and PGS002076 (Privé 2022) and PGS001338 (Tanigawa 2022). Score = Δ coefficient $\times$ dose of effect allele. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog (associations with $p \leq 5 \times 10^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** GWAS Catalog, ClinVar, dbSNP/Ensembl and reference literature: Ewing 2012 (HOXB13 G84E), Gudmundsson 2007 and Yeager 2007 (8q24), Gudmundsson 2010 and Kachuri 2023 (PSA genetic), Eeles 2008 (MSMB and KLK3), Conti 2021 and Wang 2023 (GWAS meta-analysis), Klein 2011 (SELECT trial) and the EAU monograph on screening. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Large effect variants and repair genes	HOXB13	rs138213197	C/C	chip	T	0	strand + (published alleles; absent site of VCF)	without the risk allele
Large effect variants and repair genes	CHEK2	rs17879961	A/A	chip	G	0	strand +	without the risk allele
Large effect variants and repair genes	BRCA2	rs11571833	A/A	chip	T	0	strand + (palindromic; assumed +)	without the risk allele
Large effect variants and repair genes	ATM	rs1800056	T/T	chip	C	0	strand +	without the risk allele
Common loci of prostate cancer	8q24 (2 region)	rs16901979	C/C	chip	A	0	strand +	without the risk allele
Common loci of prostate cancer	8q24 (3 region)	rs1447295	C/C	chip	A	0	strand +	without the risk allele
Common loci of prostate cancer	8q24 (1 region)	rs4242382	G/G	chip	A	0	strand +	without the risk allele
Common loci of prostate cancer	8q24 (POU5F1B/ MYC)	rs6983267	G/T	chip	G	1	strand +	1 copy of risk allele
Common loci of prostate cancer	MSMB	rs10993994	SNP not found in the data	SNP not found in the data (DR2<0.8 imputation)	T			SNP not found in the data
Common loci of prostate cancer	NKX3-1	rs1512268	T/C	chip	T	1	strand +	1 copy of risk allele
Common loci of prostate cancer	JAZF1	rs10486567	G/A	chip	G	1	strand +	1 copy of risk allele
Common loci of prostate cancer	TERT	rs2242652	G/G	chip	G	2	strand +	2 risk allele copies
Common loci of prostate cancer	CASC8 (8q24)	rs10086908	T/T	chip	T	2	strand +	2 risk allele copies
Common loci of prostate cancer	SLC22A3	rs9364554	C/T	chip	T	1	strand +	1 copy of risk allele
Genetic PSA (which changes the analysis reading)	KLK3	rs17632542	T/C	chip	C	1	strand +	T/C: less risk, but PSA naturally a little lower
Genetic PSA (which changes the analysis reading)	KLK2/KLK3	rs2735839	A/G	chip	G	1	strand +	1 copy of effect allele
Androgens and 5-alpha-reductase	SRD5A2	rs523349	C/C	chip	G	0	strand + (palindromic; assumed +)	without the protective allele (common genotype)
Androgens and 5-alpha-reductase	SRD5A2	rs9282858	C/C	chip	T	0	strand +	without the risk allele
Androgens and 5-alpha-reductase	CYP17A1	rs743572	A/G	chip	G	1	strand +	1 copy of effect allele
Androgens and 5-alpha-reductase	AR (X)	rs6152	G (X)	chip	A	0	strand +; hemizygous (chromosome X)	Allele A (X, hemizygous)

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Benign hyperplasia and inflammation	RNASEL	rs486907	C/C	imputation DR2=1.00	T	0	strand +	without the risk allele